

Convex Sets, Convex Functions and Convex Optimisation

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



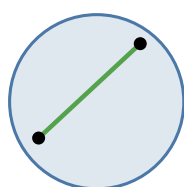
- **Builds on:** Convex and non-convex loss ([Note 590](#)); Lagrange multipliers, KKT and duality ([Note 620](#)).
- **Leads to:** Linear and quadratic programming ([Note 622](#)).

1. Overview

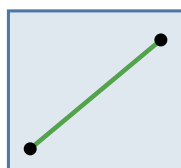
Key point: A set is convex when the segment between any two of its points stays inside it; a convex optimisation problem minimises a convex function over a convex set, and for such problems every local minimum is global and the dual gives the same answer as the primal.

This Note follows Chapter 7 (Section 7.3, opening part) of *Mathematics for Machine Learning* (Deisenroth, Faisal and Ong, 2020).

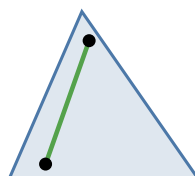
convex



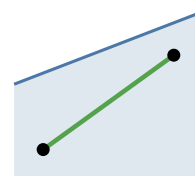
disc



box

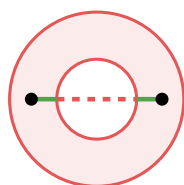


triangle

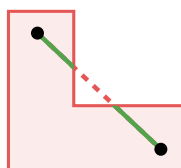


half-plane (part)

non-convex



ring



L-shape



star

dashed red: the
part of the segment
outside the set

Figure 1: Convex sets (top): every segment between two points stays inside. Non-convex sets (bottom): some segment leaves the set

Two Notes prepare this one. The [convex and non-convex cost functions Note](#) defined a convex function by the chord test and showed why gradient descent likes them. The [Lagrange multipliers Note](#) added constraints, which cut the parameter space down to a feasible region.

This Note puts the two together. Figure 1 shows the new ingredient, convex sets. Then come two practical tests for convex functions (tangent lines and the Hessian), rules for building convex functions, and the convex optimisation problem.

2. Convex sets

Key point: A set C is convex if, for any two points in it, every point on the segment between them is also in C .

2.1 The definition

Key point: The segment between $\mathbf{x}$ and $\mathbf{y}$ is all points $\theta\mathbf{x} + (1 - \theta)\mathbf{y}$ with $0 \leq \theta \leq 1$; all of them must lie in the set.

The points between $\mathbf{x}$ and $\mathbf{y}$ are written the same way as in the chord test of the [convex and non-convex cost functions Note](#): a mix $\theta\mathbf{x} + (1 - \theta)\mathbf{y}$, with θ from 0 to 1.

1. **In words:** pick any two points of the set; the whole straight segment joining them must stay in the set.
2. **Formula:** C is a **convex set** if for all $\mathbf{x}, \mathbf{y} \in C$ and all θ with $0 \leq \theta \leq 1$,

$$\theta\mathbf{x} + (1 - \theta)\mathbf{y} \in C$$

3. **Example:** the disc of radius 1, all points with length at most 1. Take $\mathbf{x} = (1, 0)$ and $\mathbf{y} = (0, 1)$, both on its edge. The midpoint ($\theta = 0.5$) is $(0.5, 0.5)$, with length $\sqrt{0.25 + 0.25} = 0.71 \leq 1$: inside.

Now the ring of points with length between 1 and 2. Take $\mathbf{x} = (1.5, 0)$ and $\mathbf{y} = (-1.5, 0)$, both in the ring. Their midpoint is $(0, 0)$, with length $0 < 1$: outside. One such pair is enough, so the ring is not convex.

Informally, a convex set has no dents and no holes (Figure 1). Common convex sets in ML:

- **a line, a plane or a hyperplane** (see the [equation of a hyperplane Note](#)), the set where a linear equation holds;
- **a half-space**, one side of a hyperplane, such as $x + y \geq 3$ in the [Lagrange multipliers Note](#);
- **a box**, such as $-1 \leq x_i \leq 1$ for every coordinate;
- **a ball** $\|\mathbf{w}\| \leq t$, the Ridge constraint, and the diamond $|w_1| + |w_2| \leq t$, the Lasso constraint.

2.2 Combining convex sets

Key point: The overlap (intersection) of convex sets is convex; their union usually is not.

If $\mathbf{x}$ and $\mathbf{y}$ lie in both A and B , the segment between them lies in A (because A is convex) and in B (because B is convex), so it lies in the overlap. This is why a feasible region made

of many convex constraints is convex: it is the overlap of one convex set per constraint. A box, for example, is the overlap of $2n$ half-spaces.

A union does not keep convexity. Two separate discs together form a set where the segment from one disc to the other crosses empty space.

Extra: The set where a convex function stays at or below a level, $\{\mathbf{x} : g(\mathbf{x}) \leq c\}$, is always convex (Boyd and Vandenberghe §3.1.6). This is why convex optimisation asks for convex functions g_i in its inequality constraints $g_i(\mathbf{x}) \leq 0$ (Section 6): each constraint then cuts out a convex set. The set where a convex function is exactly 0 is not convex in general (the circle $x^2 + y^2 = 1$ is not), which is why equality constraints must be linear.

3. Convex functions and their sets

Key point: A function is convex when every chord lies on or above its graph; equivalently, the region above its graph is a convex set.

The [convex and non-convex cost functions Note](#) defined a **convex function**: $f(\theta\mathbf{a} + (1-\theta)\mathbf{b}) \leq \theta f(\mathbf{a}) + (1-\theta)f(\mathbf{b})$ for all $\mathbf{a}$, $\mathbf{b}$ and $0 \leq \theta \leq 1$. One condition was left implicit there: the domain of f must itself be a convex set, so that the mixed point $\theta\mathbf{a} + (1-\theta)\mathbf{b}$ is somewhere f is defined.

Two related ideas:

- **Concave function:** the negative of a convex function, an upside-down bowl. Every chord lies on or below the graph. The natural logarithm is concave; so is the dual function $D(\lambda) = 3\lambda - 3\lambda^2/8$ of the [Lagrange multipliers Note](#). Maximising a concave function is the same task as minimising a convex one.
- **Epigraph:** the region on and above the graph of f , as if the bowl were filled with water. A function is convex exactly when its epigraph is a convex set (Boyd and Vandenberghe §3.1.7). This links the two halves of this Note: convex functions are convex sets seen from above.

Extra: The defining inequality is the two-point case of **Jensen's inequality**. With more points and weights $\theta_i \geq 0$ that add up to 1, a convex f satisfies $f(\sum_i \theta_i \mathbf{x}_i) \leq \sum_i \theta_i f(\mathbf{x}_i)$: the function of an average is at most the average of the function. For $f(x) = x^2$ and the points 0, 1, 2 with equal weights: $f(1) = 1 \leq (0 + 1 + 4)/3 = 1.67$. With probabilities as weights, this reads $f(E[X]) \leq E[f(X)]$ (see the [expected value and variance Note](#)); for $f(x) = x^2$ it says $E[X^2] - (E[X])^2 \geq 0$, that is, a variance is never negative.

4. Testing convexity with derivatives

Key point: For a differentiable function, convexity means every tangent line (or plane) lies on or below the graph; for a twice-differentiable one, it means the Hessian has no negative eigenvalues anywhere.

Checking every chord is impossible in practice. When derivatives exist, two shortcuts replace it.

4.1 First-order condition: tangents lie below

Key point: f is convex exactly when $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y}$: the tangent plane at any point underestimates f everywhere.

The right-hand side is the tangent plane of f at $\mathbf{x}$, the first-order Taylor approximation of the [Hessian and multivariate Taylor Note](#) (Section 4).

1. **In words:** the tangent line or plane at any point never rises above the graph.
2. **Formula:** a differentiable f is convex if and only if, for all $\mathbf{x}$ and $\mathbf{y}$ (Boyd and Vandenberghe §3.1.3),

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$$

3. **Example:** the **softplus** function $f(z) = \ln(1 + e^z)$. Its derivative is the [sigmoid](#) $\sigma(z)$, so at $z = 0$ the slope is $\sigma(0) = 0.5$ and the value is $\ln 2 = 0.693$. At $z = 2$:

$$f(2) = \ln(1 + e^2) = 2.13, \quad \text{tangent: } 0.693 + 0.5 \times (2 - 0) = 1.69$$

$2.13 \geq 1.69$: the curve is above its tangent (Figure 2, left).

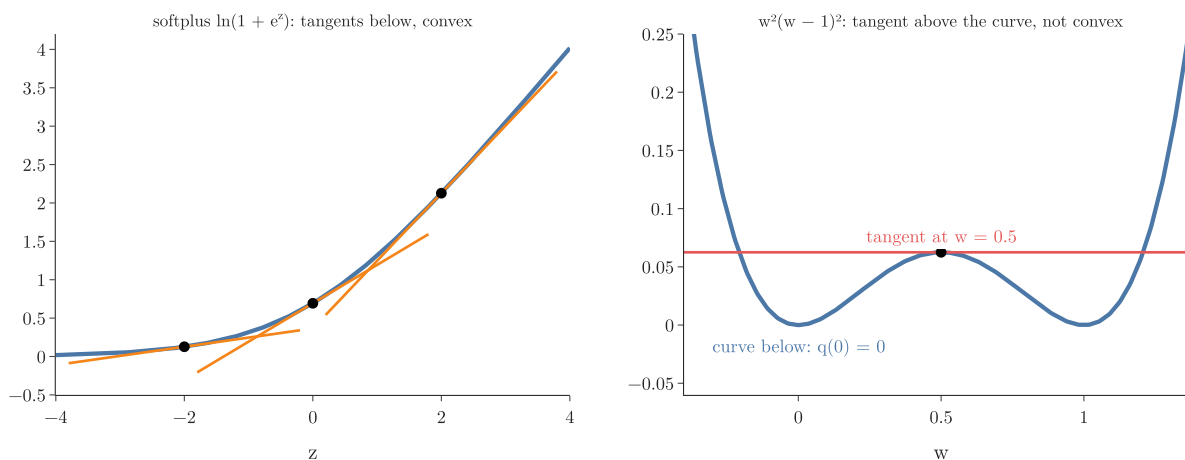


Figure 2: The first-order test. Left: every tangent of softplus lies below the curve, so it is convex. Right: the tangent of $w^2(w - 1)^2$ at $w = 0.5$ lies above the curve near 0 and 1, so it is not convex

Figure 2 (right) shows the test failing. For $q(w) = w^2(w - 1)^2$, the tangent at $w = 0.5$ is flat at height 0.0625, but $q(0) = 0$ lies below it.

The first-order condition gives the most useful fact about convex functions in one line. If $\nabla f(\mathbf{x}^*) = \mathbf{0}$, the condition becomes $f(\mathbf{y}) \geq f(\mathbf{x}^*)$ for every $\mathbf{y}$. So for a convex function, any point with zero gradient is a global minimum. Gradient descent stops at zero gradient, so on a convex function it stops at the best answer.

Extra: Softplus is the [log loss](#) in disguise. For a row with label 0 and score z , the loss $-\ln(1 - \sigma(z))$ equals $\ln(1 + e^z)$; for label 1, $-\ln \sigma(z) = \ln(1 + e^{-z})$. Both are convex in z , and $z = \mathbf{w}^\top \mathbf{x}$ is linear in the weights. A convex function of a linear function is convex (Boyd and Vandenberghe §3.2.2), so the loss of logistic regression is convex in $\mathbf{w}$: gradient descent on it reaches the global minimum.

4.2 Second-order condition: curving up everywhere

Key point: A twice-differentiable f is convex exactly when its Hessian is positive semi-definite (no negative eigenvalues) at every point; in one variable, $f''(x) \geq 0$ everywhere.

The [Hessian and multivariate Taylor Note](#) (Section 3.3) showed that the signs of the Hessian's eigenvalues give the local shape: all positive is a bowl, mixed signs a saddle. A symmetric matrix with no negative eigenvalues is positive semi-definite (see the [SVD geometry Note](#)). Convexity asks for this at every point, not just one.

1. **In words:** the function must curve upwards (or stay flat) in every direction, everywhere.
2. **Formula:**

$$f \text{ is convex} \iff \nabla^2 f(\mathbf{x}) \text{ is positive semi-definite for every } \mathbf{x}$$

In one variable: $f''(x) \geq 0$ for every x .

3. **Example:** for softplus, $f''(z) = \sigma(z)(1 - \sigma(z))$, the sigmoid's derivative (see the [sigmoid derivative Note](#)). Both factors lie between 0 and 1, so $f''(z) > 0$ everywhere: convex. At $z = 0$, $f''(0) = 0.5 \times 0.5 = 0.25$.

With two variables, compare two quadratic functions. Their Hessians are constant, so one check covers every point:

Function	Hessian	Eigenvalues	Convex?
$x^2 + xy + y^2$	$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	1 and 3	yes: a bowl
$x^2 + 3xy + y^2$	$\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$	5 and -1	no: a saddle

The chord test agrees. For $x^2 + 3xy + y^2$, the points $(1, -1)$ and $(-1, 1)$ both give $1 - 3 + 1 = -1$. Their midpoint $(0, 0)$ gives 0, above the chord value -1 .

Python: the eigenvalue check in NumPy. `eigvalsh` is meant for symmetric matrices such as Hessians and returns the eigenvalues in increasing order.

```
import numpy as np
```

```
H1 = np.array([[2, 1], [1, 2]])
```

```
H2 = np.array([[2, 3], [3, 2]])
```

```
np.linalg.eigvalsh(H1)    # [1. 3.] all >= 0: convex
```

```
np.linalg.eigvalsh(H2)    # [-1. 5.] one < 0: not convex
```

5. Building convex functions

Key point: Adding convex functions, or scaling them by non-negative numbers, keeps them convex; this is why “convex loss plus convex penalty” is convex.

In practice we rarely test a function from scratch. We build it from pieces known to be convex, using rules that keep convexity.

Non-negative weighted sums. If f_1 and f_2 are convex and $\alpha, \beta \geq 0$, then $\alpha f_1 + \beta f_2$ is convex. The proof is one step: write the chord inequality for f_1 and for f_2 , multiply each by its non-negative weight (which keeps the direction of $\leq$), and add them.

Worked with numbers: $f_1(w) = w^2$ and $f_2(w) = |w|$, with $\alpha = 1$, $\beta = 2$, between $a = -1$ and $b = 3$ at the midpoint 1:

$$\text{curve: } 1^2 + 2 \times |1| = 3, \quad \text{chord: } 0.5 \times (1 + 2) + 0.5 \times (9 + 6) = 9$$

This rule covers the regularised losses of earlier Notes:

- **Ridge:** squared error plus $\alpha \|\mathbf{w}\|^2$ (see the [Ridge regression maths Note](#)): convex plus convex.
- **Lasso:** squared error plus $\alpha \sum |w_i|$ (see the [Lasso regression Note](#)): convex plus convex, even though $|w|$ has a corner.
- **Regularised logistic regression:** log loss plus a penalty: convex plus convex.

What does not keep convexity. A difference or a product of convex functions can fail:

- **Difference:** w^2 and $2w^2$ are convex, but $w^2 - 2w^2 = -w^2$ is an upside-down bowl.
- **Product:** w^2 and $(w - 1)^2$ are convex, but their product $w^2(w - 1)^2$ has two dips (Figure 2, right). Its chord from 0 to 1 is at height 0, while the curve at 0.5 is 0.0625.

Extra: One more rule: the pointwise **maximum** of convex functions is convex (Boyd and Vandenberghe §3.2.3). The hinge loss $\max(0, 1 - z)$ of the [SVM soft margin Note](#) is the maximum of two straight lines, so it is convex, and with the convex penalty $\frac{1}{2} \|\mathbf{w}\|^2$ the soft-margin SVM is a convex problem.

6. Convex optimisation problems

Key point: Minimising a convex function subject to convex inequality constraints and linear equality constraints is a convex optimisation problem: every local minimum is global, and strong duality holds.

6.1 The definition

Key point: The objective and every inequality function must be convex, and every equality constraint must be linear (affine).

1. **In words:** a convex objective, minimised over a feasible region built from convex pieces.
2. **Formula:** the problem

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \leq 0, \quad h_j(\mathbf{x}) = 0$$

is a **convex optimisation problem** when f and every g_i are convex functions and every h_j is affine, $h_j(\mathbf{x}) = \mathbf{a}_j^T \mathbf{x} - b_j$. Then the feasible region is a convex set (Section 2.2).

3. **Example:** the problem of the [Lagrange multipliers Note](#): minimise $x^2 + 2y^2$ (Hessian with eigenvalues 2 and 4, convex) subject to $3 - x - y \leq 0$ (linear, so convex). It is a convex optimisation problem.

6.2 What convexity guarantees

Key point: No local-minimum traps, a zero-gradient (or KKT) point is the answer, and the dual reaches the same value as the primal.

For a convex optimisation problem:

- **Every local minimum is a global minimum.** The argument of the [convex and non-convex cost functions Note](#) (Section 3.2) works inside a convex feasible region too, because the segment towards a lower point stays feasible.
- **The first-order conditions are enough.** Without constraints, a point with zero gradient is the answer (Section 4.1). With constraints, a point that satisfies the KKT conditions of the [Lagrange multipliers Note](#) is the answer.
- **Strong duality.** The maximum of the dual function equals the primal minimum. In the example above, $D(4) = 6$ equals the primal minimum 6.

Extra: Strong duality for a convex problem needs one mild extra condition, **Slater's condition**: at least one point satisfies every inequality constraint strictly. For $3 - x - y \leq 0$, the point $(3, 3)$ gives $-3 < 0$, so it holds (Boyd and Vandenberghe §5.2.3).

6.3 Which ML problems are convex

Key point: The classical linear models are convex problems; trees, clustering and neural networks are not.

Problem	Convex?	Why
Linear regression , squared error	yes	Hessian $2X^T X$ is positive semi-definite

Problem	Convex?	Why
Ridge, Lasso, Elastic Net	yes	convex loss plus convex penalty
Logistic regression	yes	log loss is softplus of a linear score
SVM (hard and soft margin)	yes	convex objective, linear constraints (a quadratic program)
K-means	no	the result depends on the starting centroids
Neural networks	no	many minima and saddle points (Goodfellow et al. §8.2)

For the convex ones, the answer does not depend on the starting point or the solver, only on the data and the hyperparameters. The two best-known families of convex problems, linear and quadratic programs, are the topic of the [linear and quadratic programming Note](#).

7. Summary

Idea	Test	Example
Convex set	every segment between two points stays inside	disc yes, ring no
Convex function	chord on or above the graph	w^2 yes, $w^2(w-1)^2$ no
First-order test	tangent on or below the graph	softplus: $2.13 \geq 1.69$
Second-order test	Hessian positive semi-definite everywhere	eigenvalues 1, 3 yes; 5, -1 no
Building rules	non-negative sums and maximums keep convexity	Ridge, Lasso, hinge loss
Convex problem	convex f and g_i , affine h_j	Lagrange Note example

- Overlaps of convex sets are convex, so convex constraints give a convex feasible region.
- For a convex function, zero gradient means global minimum.
- Convex problems have no local-minimum traps and satisfy strong duality.

Sources

- Boyd, S. and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. Sections 3.1–3.2 (convex functions and the operations that keep convexity) and 5.2.3 (Slater’s condition).
- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Section 7.3 (MML).
- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. Section 8.2, challenges in neural network optimisation.

8. Key terms

Term	Meaning
Convex set	A set that contains the whole segment between any two of its points
Concave function	The negative of a convex function; every chord lies on or below its graph
Epigraph	The region on and above a function’s graph; convex exactly when the function is convex
Jensen’s inequality	For a convex function, the function of a weighted average is at most the weighted average of the function
Softplus	The function $\ln(1 + e^z)$, a smooth convex curve whose derivative is the sigmoid

Term	Meaning
First-order condition	A differentiable function is convex exactly when every tangent plane lies on or below its graph
Second-order condition	A twice-differentiable function is convex exactly when its Hessian is positive semi-definite everywhere
Convex optimisation problem	Minimising a convex function subject to convex inequality constraints and affine equality constraints
Slater's condition	Some point satisfies every inequality constraint strictly; together with convexity it guarantees strong duality